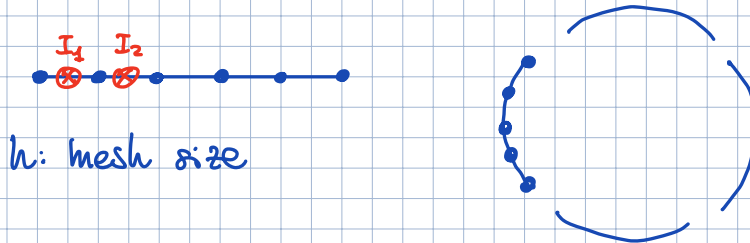


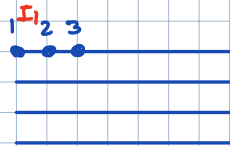
ANATOMY OF THE CODE

① Preprocessor → ② Solver → ③ Postprocessor

① 12 geometry



only DC working condition



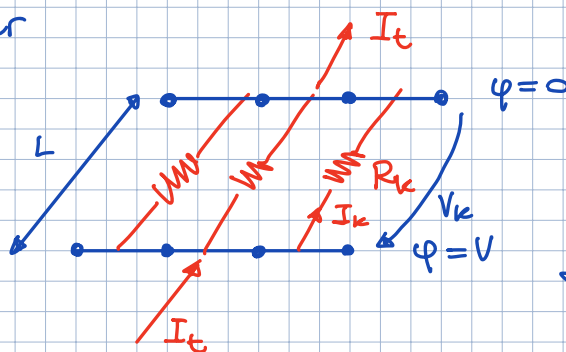
16 material's properties

$$\rho(B_n, B_t, J, T) = \frac{\epsilon_c}{J} \left(\frac{J}{J_c(B_n, B_t, T)} \right)^n$$

$10^{-4} V/m$

$$J_c(B_n, B_t, T) = \frac{J_{c0}}{\left(1 + \frac{\sqrt{\kappa^2 B_t^2 + B_n^2}}{B_0} \right)^\alpha} \frac{T_c - T}{T_c - T_{op}}$$

② solver



$$R_k = \rho_k(B_n, B_t, J, T) \frac{L}{S_k}$$

$$S_k = w_k \delta$$

$\downarrow$ mm $\downarrow$ μm

$$V_k = R_k I_k = \rho_k \frac{L}{S_k} I_k = \rho_k L J_k \quad V_k$$

$$V_1 = V_2 = \dots = V_k = V_N = V \quad \text{one unknown voltage}$$

$$\sum_{k=1}^N I_k = I_t$$

$$\begin{bmatrix} R_1 & & & & \\ & R_2 & & & \\ & & \ddots & & \\ & & & R_k & \\ & & & & \ddots \\ & & & & & R_N \\ 1 & 1 & \dots & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_k \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} V \\ V \\ \vdots \\ V \\ \vdots \\ V \\ I_t \end{bmatrix}$$

$$\begin{bmatrix} R_1(I) & & & -1 \\ & R_2(I) & & -1 \\ & & \ddots & \vdots \\ & & & R_N(I) - 1 \\ 1 & 1 & \dots & 1 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \\ V \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ I_t \end{bmatrix}$$

$$A(x) x = b$$

2b. additional functions

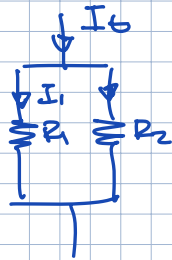
Biot-Savart-like machinery

2c. solution

FSOLVE

$$F(x) = 0$$

$$F(x) = A(x)x - b$$



$$I_1 = \frac{R_2}{R_1 + R_2} I_t$$

$$I_1 = \frac{G_1}{G_1 + G_2} I_t$$

$$I_1 = \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}} I_t$$



3 uuk
6 uuk

$V^{(1)}$
 $V^{(2)}$

$$\varepsilon^{(2)} = \frac{V^{(2)} - V^{(1)}}{V^{(2)}}$$

